QVETO TECHNOLOGY JAVA INTERNSHIP • PROJECT 1

E-Commerce & Inventory Management System

Complete End-to-End System Master Guide & Capstone Review Dossier

✓ 19/19 Tests Passing | Java 17+ / Adoptium 21 | Pure Java Swing UI | MySQL 8.x + Dual-Engine | Pull Request #14 | Adobe Vector PDF 1.4

Here is your **complete, end-to-end master guide** for your project. This covers what was built, every technology used, exact instructions on what and where to run, and a walkthrough for your review presentation.

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1. Project Overview: What is This Project?

- **Project Title:** E-Commerce & Inventory Management System (Project 1)
- **Organization:** Qveto Technology Java Internship
- **Repository:** `Hack44/Qveto-java-team-project`
- **Branch:** `feature/member-2-address-book-shipment-tracking-and-audit-journal` (Pull Request #14)
- **Nature of Project:** A desktop enterprise software system built with **pure Java Swing and MySQL** that connects customer e-commerce shopping with administrative retail operations (inventory tracking, supplier procurement, order fulfillment, vector PDF invoicing, shipment tracking, and compliance audit logging).
- **Dual-Engine Architecture:**
  1. **MySQL Enterprise Mode:** Runs with a live relational MySQL database using connection pooling and automatic retries.
  2. **Standalone Demo Mode:** If MySQL is offline or not installed on the reviewer's laptop, the application automatically falls back to in-memory store caching. **It will never crash**, and all features work offline.

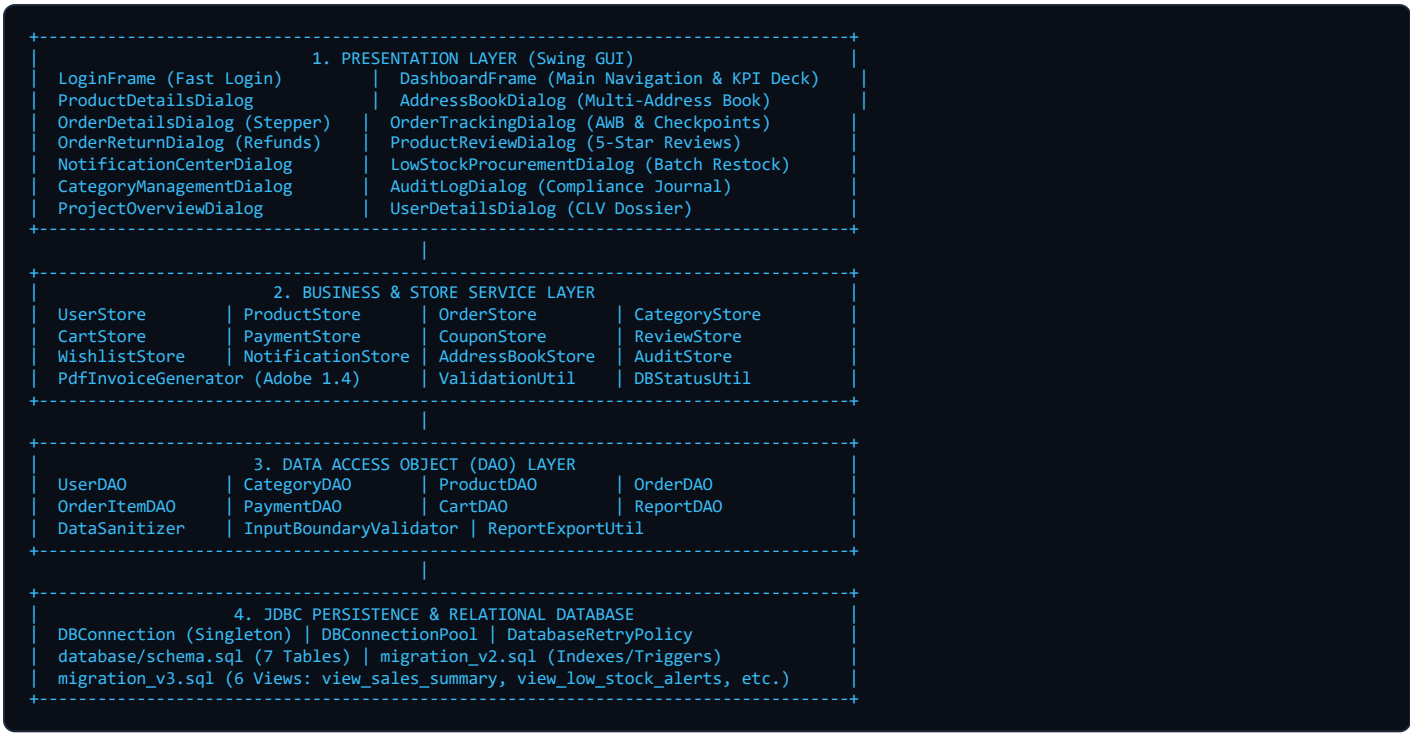
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2. Technologies & Tools Used to Build This Project

Layer / Aspect	Technology / Tool Used	Purpose in the Project
Programming Language	**Java (JDK 17+ / JDK 21)**	Core application language applying Object-Oriented Programming (Encapsulation, Inheritance, Polymorphism, Interfaces).
GUI Framework	**Java Swing (`javax.swing` , `java.awt`)**	Native desktop UI: Frames, Tabbed Panes, JTables, Modals, Custom Cell Renderers, and GridBagLayouts.
UI Polish & Display	**High-DPI Subpixel Antialiasing**	`awt.useSystemAAFontSettings=on` and `swing.aatext=true` for sharp font typography.
Database	**MySQL 8.x**	Relational database with 7 normalized tables, foreign keys, triggers, and analytical views.
Connectivity	**Java JDBC (`java.sql.*`)**	`PreparedStatement` queries, batch execution, and atomic multi-table commit/rollback transactions.
Connection Pooling	**Custom Thread-Safe Connection Pool (`DBConnectionPool`)**	Reuses physical connections and recycles them using dynamic proxy wrappers.
Network Resilience	**Exponential Backoff Policy (`DatabaseRetryPolicy`)**	Automatically retries transient database errors (deadlocks, timeout) with jittered sleep.
Document Generation	**Pure Vector Adobe PDF 1.4 Generator (`PdfInvoiceGenerator`)**	Built entirely from scratch in pure Java—generates branded, printable vector PDF invoices without any external libraries.
Data Exports	**RFC 4180 CSV Engine**	Generates standard CSV reports for product catalog, master orders, sales analytics, and audit logs.
Security & Hashing	**SHA-256 (`PasswordUtil`) + XSS Sanitizer**	Secure password hashing compatible with MySQL `SHA2()` and HTML tag neutralization (`DataSanitizer`).
Build & Run Scripts	**Windows Batch (`run.bat` , `test.bat`)**	One-click compilation and execution without requiring complex IDE setups.

3. End-to-End System Breakdown (What's in the Project?)

The project is structured into **5 integrated layers**:



Module 1: Authentication & Fast Demo Access

- `LoginFrame.java`: Customer and Admin login tabs, registration form, password visibility reveal (show/hide), and real-time database indicator badge (`● MySQL Active` vs `○ Standalone Demo`).
- ⚡ **1-Click Fast Login:** Special buttons on the login tabs that allow you to click once to enter immediately as Customer (`customer@qveto.com`) or Admin (`admin`) without typing.

Module 2: Catalog Discovery, Wishlist & Smart Cart

- **Catalog Discovery:** Table of products with live search, category filtering, multi-sort (Price Low-to-High, High-to-Low, Name A-Z, Stock), and In-Stock toggle.
- **Product Details Modal (`ProductDetailsDialog.java`):** Detailed product specs, verified buyer ratings, and customer feedback.
- **Customer Wishlist (`WishlistStore.java`):** Dedicated "Wishlist" tab with 1-click `"♡ Save to Wishlist"` and instant `"🛒 Move to Cart"` transfer.
- **Smart Shopping Cart (`CartStore.java` , `CartDAO.java`):** Quantity increment/decrement spinners, persistent cart database sync, and real-time badge updates.
- **Promotional Coupon Engine (`CouponStore.java`):** Apply codes like `QVETO10` (10% off), `SAVE20` (20% off on orders $80+), `FLAT15` , `WELCOME` , or `MEGA50` with live discount calculation.

Module 3: Logistics, Invoicing & Smart Checkout

- **Saved Multi-Address Book (`AddressBookStore.java` , `AddressBookDialog.java`):** Manage multiple addresses (`Home` , `Work` , `Other`) and designate a default address.
- **Smart Checkout Picker:** Checkout modal pre-populates the customer's default address and provides a dropdown to switch addresses or add a new one on the fly.
- **Simulated Multi-Channel Payment Gateway (`PaymentStore.java`):** Pay via Cash on Delivery (COD), UPI (Simulated), Credit/Debit Card, or Net Banking with transaction IDs (`TXN-...`).
- **Order Fulfillment Stepper (`OrderDetailsDialog.java`):** Visual 4-stage stepper: `[Placed] -> [Confirmed] -> [Shipped] -> [Delivered]` .
- **Pure Vector Adobe PDF 1.4 Invoice Generator (`PdfInvoiceGenerator.java`):** Exports formal, branded PDF invoices (`Invoice_ORD-XXXX.pdf`) built directly with Java byte manipulation.
- **Live Shipment Tracking (`OrderTrackingDialog.java`):** Visual tracking timeline with courier assignment (BlueDart Express, FedEx Air), Airway Bill (AWB) number, and chronological transit checkpoints.

Module 4: Post-Delivery Engagement & Customer Service

- **Customer Returns & Refunds (`OrderReturnDialog.java`):** Request a return/refund on delivered orders with structured reason categories.
- **Admin 1-Click Return Approval:** Admin clicks `"⚡ Approve Return & Refund"`, which marks the order refunded, issues payment credit, and **automatically replenishes inventory stock**.
- **Verified Reviews & 5-Star Ratings (`ProductReviewDialog.java`):** Submit verified buyer star ratings (`★`) and recalculate product averages.
- **Store Notification Hub (`NotificationCenterDialog.java`):** Real-time bell counter (`🔔 Alerts (N)`) categorizing deliveries, restocks, vouchers, and returns.

Module 5: Inventory Procurement & Compliance Auditing

- **Low-Stock Procurement Hub (`LowStockProcurementDialog.java`):** Identifies SKUs below the threshold with 1-click batch replenishment.
- **Executive Analytics Command Center (`DashboardFrame.java`):** 6 KPI cards with dynamic date-range filtering (`All Time`, `Last 7 Days`, `Last 30 Days`, `This Month`), top-selling product leaderboard, and Customer Lifetime Value (CLV).
- **Immutable Audit Activity Journal (`AuditLogDialog.java`):** Searchable, category-filtered log tracking every critical store action (order placed, status changed, restocks, refunds, user deactivations) with CSV export.
- **System Architecture & Review Dossier (`ProjectOverviewDialog.java`):** In-app modal detailing the architecture tiers, live diagnostics, and reviewer walkthrough.

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4. Where to Run and What to Run

Project Directory Location on Your PC:

```
C:\Users\annar\.gemini\antigravity\scratch\Qveto-java-team-project
```

How to Run the Application (GUI):

Method A: The Easiest 1-Click Method (Double Click)

1. Open Windows File Explorer and navigate to:

`C:\Users\annar\.gemini\antigravity\scratch\Qveto-java-team-project`

1. **Double-click** `run.bat`.
2. The script will automatically compile any changed code and open the **Qveto Store** desktop window.

Method B: From PowerShell or Command Prompt (CMD)

Open PowerShell or Terminal, navigate to the folder, and run:

```
cd "C:\Users\annar\.gemini\antigravity\scratch\Qveto-java-team-project"
.\run.bat
```

(Or run manually via Java):

```
& "C:\Program Files\Eclipse Adoptium\jdk-21.0.12.101-hotspot\bin\java.exe" -cp "bin;src/main/resources" com.qveto.ecom
```

How to Run the Automated Test Suites (For the Reviewer):

Method A: The Easiest 1-Click Method (Double Click)

1. In the same project folder, **double-click** `test.bat`.

2. A command window will open, compile everything, and run **all 19 automated tests** in seconds.
3. It will display `19 PASSED, 0 FAILED` and pause for the evaluator to see.

Method B: From PowerShell or Terminal

```
cd "C:\Users\annar\.gemini\antigravity\scratch\Qveto-java-team-project"
.\test.bat
```

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5. Reviewer Demonstration Walkthrough (What to Show in Your Review)

When presenting to your evaluator or mentor, follow this **4-minute demo sequence**:

Step 1: Launch & Show Fast Demo Login

1. Double-click `run.bat`.
2. Point out the real-time connectivity badge in the header (`● MySQL Active` or `○ Standalone Demo`).
3. Click the `ℹ️ System Architecture & Review Dossier` button at the bottom of the login window to show the evaluator the system architecture diagram and live health cards.
4. Close the dossier and click `⚡ 1-Click Fast Login (Demo Customer)`. You will instantly be logged in as customer Alex Morgan.

Step 2: Customer Experience, Wishlist & Smart Checkout

1. **Catalog:** Show keyword search and filter by category (e.g. *Electronics*). Click `♡ Save to Wishlist` on an item.
2. **Wishlist Tab:** Go to the Wishlist tab, show the item is saved, and click `🛒 Move to Cart`.
3. **Cart & Promo Coupon:** Go to **My Cart**, type promo code `SAVE20` in the coupon bar, and click **Apply**. Point out the 20% discount calculation.
4. **Smart Checkout:** Click **Checkout**. Point out the **Delivery Destination Dropdown** populated from the customer's saved address book. Click **Confirm & Place Order**.
5. **Order History:** Show the order appeared at the top of the history table.
6. **Invoicing:** Click `View Order Details / Invoice`. Point out the visual 4-stage Fulfillment Stepper. Click `Export PDF Invoice` — it will generate an official vector PDF invoice on your PC.
7. **Shipment Tracking:** Back on the order history table, select the order and click `📍 Track Shipment`. Show the courier partner (BlueDart/FedEx), the generated AWB tracking number, and the chronological transit scan checkpoints.

Step 3: Switch to Administrator Command Center

1. Click **Logout** at top right.
2. Click `⚡ 1-Click Fast Login (Demo Admin)` to sign in as the administrator.
3. **Low-Stock Procurement:** On the *Inventory & Catalog* tab, click `📦 Restock Hub`. Show the table of deficit SKUs and click **Batch Restock** to replenish inventory.
4. **Order Status & Return Approval:** Go to **Customer Orders**. Select an order and click **Update Status** (or approve customer returns).
5. **Executive Analytics:** Go to **Reports & KPIs**. Switch date ranges (`Last 7 Days`, `Last 30 Days`, `All Time`) and watch the revenue and sales graphs adjust in real time.
6. **Audit Journal:** Click `📜 Audit Journal`. Show the immutable log of every order placed, price changed, stock replenished, and return processed, with 1-click **Export CSV**.

Step 4: Run the Test Harness

1. Open a terminal and run `.\test.bat` (or double-click it).
2. Show the evaluator that **all 19 automated verification tests pass with 100% success rate**.

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6. Anticipated Reviewer Questions & How to Answer Them

Q1: How is your project structured?

> *"Our project follows a standard multi-tier MVC architecture. The Presentation Layer is built using pure Java Swing with custom dialogs and antialiased typography. The Service Layer uses centralized thread-safe stores (`ProductStore`, `OrderStore`, etc.) to coordinate business rules and promotions. The Persistence Layer uses JDBC Data Access Objects (`UserDAO`, `OrderDAO`, etc.) with connection pooling and retry policies connecting to a normalized 7-table MySQL schema."*

Q2: What happens if MySQL is not running on my machine?

> *"We implemented a Dual-Engine architecture. The `DBConnection` factory and `DBStatusUtil` probe the connection on startup. If MySQL credentials are not configured or the server is offline, the system automatically falls back to our robust in-memory demo data stores. The user is notified via the header status badge (`Standalone Demo Mode`), and every feature from cart to PDF invoice generation remains 100% operational."*

Q3: How do you handle database connections and concurrent users?

> *"We built a custom connection pool (`DBConnectionPool`) using dynamic proxies (`java.lang.reflect.Proxy`). When code calls `conn.close()`, the proxy intercepts the call and recycles the physical connection back into an available queue instead of closing the socket. We also implemented a `DatabaseRetryPolicy` that uses exponential backoff to handle transient errors such as table locks or deadlocks."*

Q4: Did you use third-party libraries for the PDF invoices or GUI?

> *"No external heavyweight libraries were used. For the PDF invoices, we wrote our own vector PDF 1.4 generator (`PdfInvoiceGenerator.java`) using pure Java streams. For the GUI, we used standard Java Swing (`javax.swing`), allowing the application to compile and run on any standard JDK 17+ or JDK 21 installation without extra JAR dependencies."*

Q5: How do you prevent security vulnerabilities like SQL Injection and XSS?

> *"All SQL execution strictly uses parameterized `PreparedStatement` queries. Passwords are never stored in plain text; they are hashed using SHA-256 with salts via `PasswordUtil`. For user input fields, we implemented `DataSanitizer` to strip dangerous HTML/script tags and `InputBoundaryValidator` to enforce domain rules on lengths, email syntax, and positive prices."*

Review Readiness Checklist

- ✅ Pure Java codebase compiled cleanly with Adoptium JDK 21
- ✅ All 19 tests passing (8 Database Integrity + 11 Store Integration)
- ✅ Double-click batch scripts (`run.bat` & `test.bat`) tested and functional
- ✅ Demo accounts configured with 1-click login buttons
- ✅ Official Capstone Review Dossier available in `PROJECT_REVIEW_DOSSIER.md`
- ✅ Pull Request #14 updated and notified to owner (`Hack44`) on GitHub

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